

Analog IC Design (Xschem, Ngspice, ADT)**Lab 04****Common Drain Amplifier****Intended Learning Objectives**

In this lab you will:

- Learn how to use the Sizing Assistant (SA) to size the transistors.
- Design and simulate a common-drain amplifier.
- Use PMOS input transistor to avoid body effect in a CD amplifier.
- Investigate the ringing and peaking problem in a capacitive-loaded CD amplifier with a large source resistance (R_{sig}) and learn how to solve it.

Part 1: Device Sizing Using SA

- 1) From the square law, we have

$$g_m = \frac{2I_D}{V_{ov}} \rightarrow V_{ov} = \frac{2}{g_m/I_D}$$

For a real MOSFET, if we compute V_{ov} and $\frac{2}{g_m/I_D}$ they will not be equal. We can define V^* (V^*), which is calculated from actual simulation data using the formula

$$V^* = \frac{2}{g_m/I_D} \leftrightarrow g_m = \frac{2I_D}{V^*}$$

The lower the V^* the higher the g_m , but the larger the area and the lower the speed. An often used sweet-spot that provides good compromise between different trade-offs is $V^* = 200mV$.

- 2) Although the V^* is a nice parameter that is inspired by the square-law, it does not have an intuitive or a physical meaning (it is not an actual voltage in the circuit). We actually defined V^* in order to be able to define a relation between the g_m and I_D . Thus, the real parameter that we should care about is the g_m over I_D ratio (g_m/I_D).

If the square-law is valid

$$g_m = \frac{2I_D}{V_{ov}} \rightarrow \frac{g_m}{I_D} = \frac{2}{V_{ov}}$$

Using V^*

$$\frac{g_m}{I_D} = \frac{2}{V^*}$$

A small g_m/I_D means large V_{ov} (biasing in strong inversion) and a large g_m/I_D means small V_{ov} (biasing in weak inversion).

- 3) There are many good things about using the g_m/I_D as a design knob:

- The g_m/I_D gives a direct relation between the most important MOSFET parameter (g_m) and the most valuable resource (I_D). For example, a $g_m/I_D = 10 \text{ S/A}$ means you get $10 \mu\text{S}$ of g_m for every $1 \mu\text{A}$ of bias current.
 - The g_m/I_D is a normalized knob: it has a limited search range (typically from 5 to 25 S/A) independent of the technology or the device type.
 - The g_m/I_D is intuitive because it tells you directly about the inversion level (bias point) and consequently all related trade-offs. For example, $g_m/I_D = 5 \text{ S/A}$ means strong inversion (SI), $g_m/I_D = 15 \text{ S/A}$ means moderate inversion (MI), and $g_m/I_D = 25 \text{ S/A}$ means weak inversion (WI).
 - The g_m/I_D is an orthogonal knob: If we define the g_m/I_D then we define the inversion level (bias point). If you change I_D or L while keeping g_m/I_D fixed, then the inversion level (bias point) is kept fixed. The W is treated as an output variable instead of being treated as an input variable.
 - The higher the g_m/I_D (the lower the V^*) the higher the efficiency and the headroom (the available swing), but the larger the area and the lower the speed. An often used sweet-spot that provides good compromise between different trade-offs is $g_m/I_D = 10 \text{ S/A}$ ($V^* = 200\text{mV}$). Larger g_m/I_D is usually used for low-voltage and low-power designs.
- 4) We want to design a CD amplifier that has ideal current source load with the parameters below.

Parameter	
Input transistor	PMOS
L^1	$1\mu\text{m}$
V^*	200mV
Quiescent (DC) input voltage	0V
Supply	1.8V
Current consumption	$10\mu\text{A}$

- 5) We assume we use a PMOS transistor that is placed in a dedicated n-well to be able to connect the body and source terminals. This will avoid the degradation of the CD amplifier gain due to body effect.
- 6) Since the square-law is not accurate, we cannot use it to calculate the sizing. Instead, we will use the Sizing Assistant (SA) which is a powerful analog calculator that uses LUTs that are pre-generated from the simulations. The input of SA is shown below. Note that since both the gate and the drain are connected to ground, we can set $V_{DS} = V_{GS}$.

¹ A relatively long L is intentionally used to be able to use the simplified frequency response expressions given in the lecture slides.

VTH
VDSAT
GM
GDS
GMB
CDB
CGD
CGS
CSB

3) Check that the transistor operates in saturation.

➔ Hint: Example ngspice code:

```
.control

alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset

save all
save @m.xml.m0[id]
save @m.xml.m0[vgs]
save @m.xml.m0[vds]
save @m.xml.m0[vdsat]
save @m.xml.m0[vth]
save @m.xml.m0[gm]
save @m.xml.m0[gds]
save @m.xml.m0[gmb]
save @m.xml.m0[cgd]
save @m.xml.m0[cdd]
save @m.xml.m0[css]
op
write cd_amp_ac.raw
.endc
```

➔ Hint: An alternative way is to print the OP parameters of all transistors in ngspice terminal (save command is NOT required). Example (you can add additional parameters):

```
show m : id : gm : gds : gmb : vgs : vth : vds
```

2. AC Analysis

- 1) Perform AC analysis (1Hz:10GHz, logarithmic, 20points/decade) to investigate the frequency domain peaking.
- 2) Report the Bode plot magnitude.
- 3) Do you notice frequency domain peaking? How much is the peaking?
- 4) Analytically calculate the quality factor (use approximate expressions). Is the system underdamped or overdamped?
- 5) Perform parametric sweep: $C_L = 2\text{p}, 4\text{p}, 8\text{p}$.
 - Report Bode plot magnitude overlaid on same plot.
 - Report the peaking vs C_L .
 - Comment on the results.
- 6) Perform parametric sweep: $R_{sig} = 20\text{k}, 200\text{k}, 2\text{M}$.

- Report Bode plot magnitude overlaid on same plot.
- Report the peaking vs R_{sig}.
- Comment on the results.

➔ Hint: Example ngspice code (courtesy of Eng. Abdelrahman Ali):

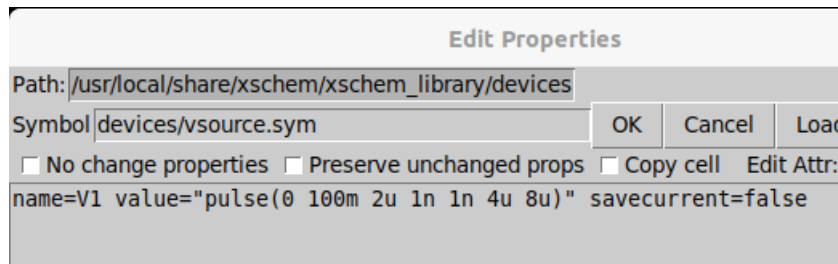
```
.control

alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset

set outfile = "peaking_results.txt"
save all
save @m.xml.m0[id]
save @m.xml.m0[vgs]
save @m.xml.m0[vds]
save @m.xml.m0[vdsat]
save @m.xml.m0[vth]
save @m.xml.m0[gm]
save @m.xml.m0[gds]
save @m.xml.m0[gmb]
save @m.xml.m0[cgs]
save @m.xml.m0[cgd]
save @m.xml.m0[csb]
save @m.xml.m0[cdb]
op
write cd_amp_ac.raw
ac dec 20 1 10g
set appendwrite
write cd_amp_ac.raw
echo "\"CD Amplifier AC Sim Results\" > $outfile
foreach cap 2p 4p 8p
    echo "\"Running Sim using Cap=$cap\"
    alter cl=$cap
    ac dec 20 1 10g
    write cd_amp_ac.raw
    meas ac gain_peak_cap MAX vdb(vout)
    echo "\"CL = $cap, Gain Peak (dB) = $&gain_peak_cap \" >> $outfile
end
* Reset C1 before R sweep
alter cl=2p
foreach my_res 20k 200k 2meg
    echo "\"Running Sim using Res=$my_res\"
    alter rl=$my_res
    ac dec 20 1 10g
    write cd_amp_ac.raw
    meas ac gain_peak_res MAX vdb(vout)
    echo "\"RES = $my_res, Gain Peak (dB) = $&gain_peak_res \" >>
$outfile
end
.endc
```

3. Transient Analysis

- 1) Use a pulse source as your transient stimulus and set it as follows: delay time = 2us, initial (zero value) = 0V, period = 8us, pulse (one value) = 100mV, fall time = 1ns, rise time = 1ns, pulse width = 4us.



- 2) Run transient analysis for 10us with 10ns step to investigate the time domain ringing.
- 3) Report VSIG and VOUT overlaid vs time.
- 4) Calculate the DC voltage difference (DC shift) between Vin and Vout.
 - What is the relation between the DC shift and VGS of the transistor?
 - How to shift the signal down instead of shifting it up?
- 5) Do you notice time domain ringing? How much is the overshoot?
- 6) Perform parametric sweep: CL = 2p, 4p, 8p.
 - Report Vout vs time overlaid on same plot.
 - Report the overshoot vs CL.
 - Comment.
- 7) Perform parametric sweep: Rsig = 20k, 200k, 2M.
 - Report Vout vs time overlaid on same plot.
 - Report the overshoot vs Rsig.
 - Comment on the results.

➔ Hint: Example ngspice code (courtesy of Eng. Abdelrahman Ali):

```
.control

alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset

set outfile = "overshoot_results.txt"
tran 10n 10u
write cd_amp_tran.raw
set appendwrite
echo "\"CD Amplifier Transient Sim Results\" > $outfile
foreach cap 2p 4p 8p
    echo "\"Running Sim using Cap=$cap\"
    alter cl=$cap
    tran 10n 10u
    write cd_amp_tran.raw
    meas tran vpeak MAX v(vout)
    meas tran vfinal AVG v(vout) FROM=5.9u TO=6u
    meas tran vinitia AVG v(vout) FROM=1u TO=1.1u
    let overshoot_percent=(vpeak - vfinal)/(vfinal-vinitia) * 100
    echo "\"CL = $cap, Overshoot (%) = $&overshoot_percent\" >>
$outfile
```

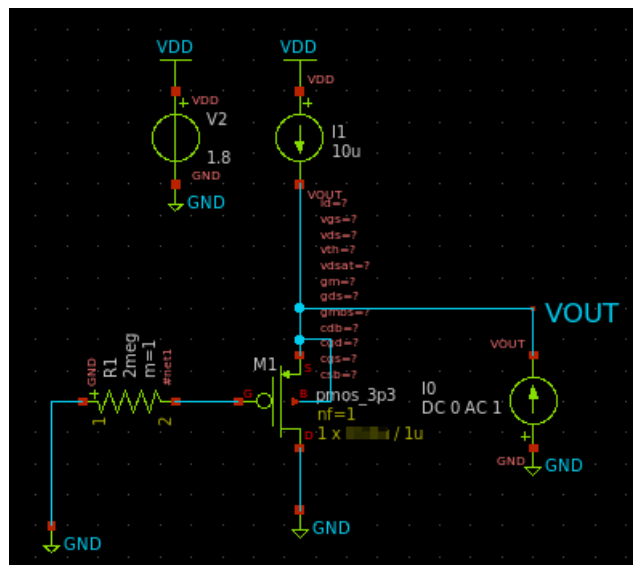
```

end
alter cl=2p
foreach my_res 20k 200k 2meg
    echo "\"Running Sim using Res=$my_res\"
    alter r1=$my_res
    tran 10n 10u
    write cd_amp_tran.raw
    meas tran vpeak MAX v(vout)
    meas tran vfinal AVG v(vout) FROM=5.9u TO=6u
    meas tran vinitial AVG v(vout) FROM=1u TO=1.1u
    let overshoot_percent = (vpeak - vfinal)/(vfinal - vinitial) * 100
    echo "\"Rsig = $my_res, Overshoot (%) = $&overshoot_percent\" >>
$outfile
end
.endc

```

4. Z_{out} (Inductive Rise)

- 1) We want to simulate the CD amplifier output impedance. Replace CL with an AC current source with magnitude = 1. Remove the AC input signal.



- 2) Perform AC analysis (1Hz:10GHz, logarithmic, 10points/decade). The voltage across the AC current source is itself the output impedance.
- 3) Plot the output impedance (magnitude and phase) vs frequency. Do you notice an inductive rise? Why?
- 4) Does Z_{out} fall at high frequency? Why?
Hint: C_{gd} appears in parallel with R_{sig} .
- 5) Analytically calculate the zeros, poles, and magnitude at low/high frequency for Z_{out} .
- 6) From the Bode plot, find the corresponding simulation values. Compare the results in a table.

➔ Hint: Example ngspice code (courtesy of Eng. Abdelrahman Ali):

```

.control

alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset

```

```

set outfile = "zout_results.txt"
ac dec 10 1 10g
write cd_amp_zout.raw
echo "\"CD Amplifier Zout Sim Results\" > $outfile
meas ac zout_low_freq FIND vmag(vout) AT=1k
meas ac zout_high_freq MAX vmag(vout)
echo "\"Zout @ 1kHz      = $&zout_low_freq\" >> $outfile
echo "\"Zout peak (max)   = $&zout_high_freq\" >> $outfile
.endc

```

5. [Two Bonus Points] How to solve the peaking/ringing problem?

- 1) The simple solution is to place the input/output poles away from each other (as we did when we swept CL and Rsig).
 - 2) Alternatively, a compensation network can be used to compensate for the negative input impedance and prevent overshoots. Read [Johns and Martin, 2012] Section 4.4 and try to implement the compensation network.
- ➔ [Johns and Martin, 2012] Note: If the source-follower buffer is intended to be used in an op-amp (and thus feedback will be placed around the buffer), and if the resonant frequency of the source-follower is substantially greater than the unity-gain frequency of the amplifier, then the overshoot can be tolerated and no compensation network is necessary.

Lab Summary

In Part 1 you learned:

- How to find transistor sizing using the Sizing Assistant (SA).
- How to design a PMOS common-drain amplifier.

In Part 2 you learned:

- How to perform AC, DC and transient simulations of a CD amplifier.
- How the peaking in the frequency response of a CD amplifier changes with the load capacitor and source resistance.
- How the ringing in the transient response of a CD amplifier changes with the load capacitor and source resistance.
- How the output impedance of a CD amplifier shows inductive behavior.

Acknowledgements

Thanks to all who contributed to these labs. Special thanks to Dr. Sameh A. Ibrahim for reviewing and editing the labs. If you find any errors or have suggestions concerning these labs, please contact Hesham.omran@eng.asu.edu.eg.